

Q13 Ans: B

Using $d \sin \theta = n\lambda$

$$d = \frac{(3)(5.2 \times 10^{-7})}{\sin(51.3^\circ)} = 2.0 \times 10^{-6} \text{ m}$$

$$\text{number of lines per mm} = \left(\frac{1}{2.0 \times 10^{-6}} \right) \times 10^{-3} = 500$$

Q14 Ans: C